1. Which address(es) should be allowed to mint/burn the LP tokens?

The only address that should be allowed to mint or burn LP tokens is the DEX smart contract address itself.

2. In what way do DEXs level the playing ground between a powerful and resourceful trader (HFT/institutional investor) and a lower resource trader (retail investors, like you and me!)?

In traditional finance, only institutional high-frequency traders (HFTs) with millions in capital and direct exchange server access can act as market makers and earn trading fees. In a DEX, absolutely anyone can deposit a few dollars of assets into the smart contract to become a Liquidity Provider (LP). This allows retail investors to earn a proportional slice of the 0.3% trading fees right alongside institutional whales. Furthermore, the x.y = k pricing algorithm is 100% transparent and deterministic, meaning institutions cannot use opaque order books or hidden liquidity to manipulate pricing.

3. Suppose there are many transaction requests to the DEX sitting in a miner's mempool. How can the miner take undue advantage of this information? Is it possible to make the DEX robust against it?

A miner can take advantage of this by using Miner Extractable Value (MEV), which happens through methods like "front-running" or a "sandwich attack". Since miners can see transactions that are waiting to be added to a block, they can spot a big buy order from a regular user that will make the price of Token A go up. The miner can then add their own buy order right before the big order and a sell order right after. This lets them make a profit from the price change caused by the regular user. To make a DEX safer against this, it can use a "Minimum Output Amount" setting, also known as Slippage Tolerance. The user signs a message showing the least amount of tokens they are okay with getting. If a miner tries to change the pool's balance so the user gets less than their minimum, the smart contract cancels the whole transaction.

4. We have left out a very important dimension on the feasibility of this smart contract- the gas fees! Every function call needs gas. How does gas fees influence economic viability of the entire DEX and arbitrage?

Gas fees add a fixed cost to every transaction, and this cost depends on the network being used. This fee has nothing to do with how big the trade is.

For the DEX: This makes very small transactions too expensive.If a user wants to swap $10 worth of tokens but the gas fee on the Ethereum network is $15, they probably won't use the DEX at all.

For Arbitrage: It creates a clear minimum profit requirement.An arbitrage trader will only make a trade if the profit from the price difference is more than the total cost of the 0.3% DEX fee plus the gas fees needed to make several smart contract calls on different exchanges.

5. Could gas fees lead to undue advantages to some transactors over others? How?

Yes, gas fees favor big players like "whales" and large institutions more than regular users.

• Proportional Cost: If a $50 gas fee is applied to a $100 trade by a regular user, it's like paying a 50% tax.But the same $50 fee on a $1,000,000 trade by an institution is just a tiny 0.005% tax.

• Priority Gas Auctions: Rich users can pay a huge extra amount to network validators to get their transactions processed first.This lets them speed up their trades, possibly take advantage of other users' trades, and use strategies like MEV that smaller traders can't afford to compete with.

6. What are the various ways to minimize slippage (as defined in 4.2) in a swap?

There are three main methods to reduce slippage in a constant product AMM:

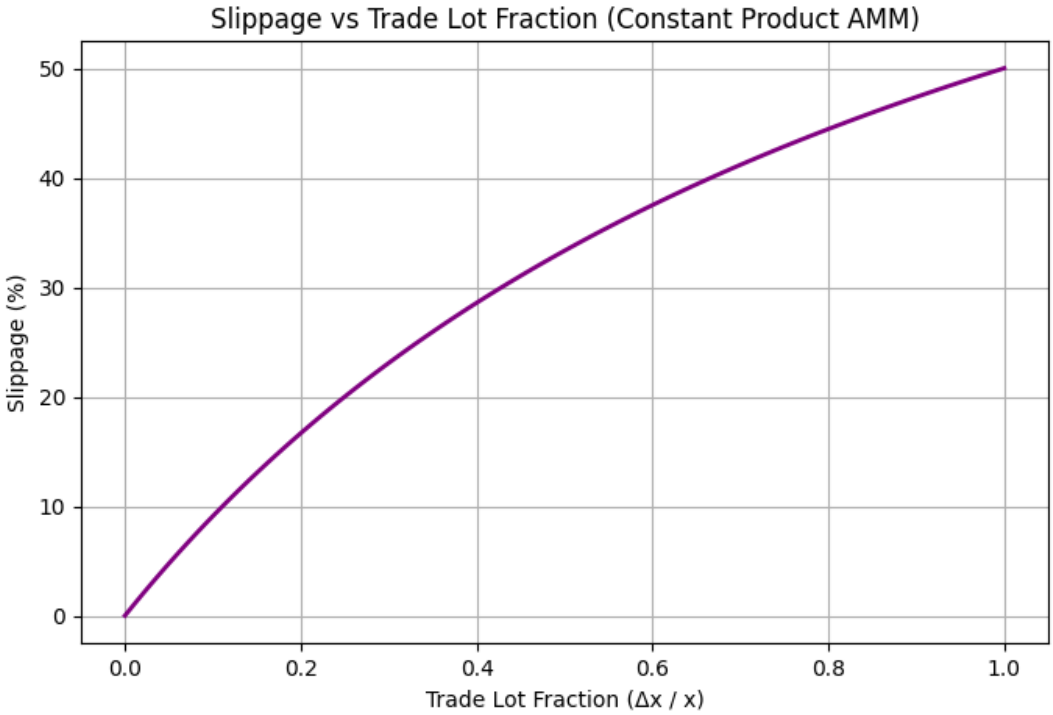
1. Deeper Liquidity: The most straightforward mathematical approach. A higher constant product ($k$) flattens the pricing curve, resulting in smaller percentage changes in reserves for larger trades.

2. Order Splitting: Instead of making one large trade that leads to significant slippage, a user can split their trade into smaller portions over time, enabling arbitrageurs to rebalance the pool between swaps.

3. Smart Routing: Advanced aggregators distribute a single large trade across multiple DEXs simultaneously, drawing from the shallowest parts of various liquidity pools rather than depleting a single one.

7. Having defined what slippage, plot how slippage varies with the "trade lot fraction" for a constant product AMM. *Trade lot fraction ($f$) is the ratio of the amount of token X deposited in a swap, to the amount of X in the reserves just before the swap* *($f = \frac{\Delta x}{x}$).*

By definition, Slippage ($S$) is the relative difference between the Actual Price and Expected Price. For a pure constant product AMM (ignoring fees).


Slippage vs Trade Lot Fraction (Constant Product AMM)
Slippage (%)
Trade Lot Fraction (Δx / x)
0
10
20
30
40
50
0.0
0.2
0.4
0.6
0.8
1.0